MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, MAGNETRON

## TYPE 6177A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

**DESCRIPTION:** CW, integral magnet, mechanically modulated, tuned frequency 4,268 to 4,350 MHz

**ABSOLUTE RATINGS:**

| Parameter: | Ef | tk | Reed drive | Reed drive | FM deviation | Eb | Ib | Ehk | Pi | Anode T | VSWR |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | sec | V | W | MHz | V | mAdc | V | W | °C | Ratio |
| Maximum: | 7.0 | --- | --- | 0.55 | 87 | 350 | 35 | 45 | 12 | 110 | 2:1 |
| Minimum: | 5.7 | 40 | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| | (Note 2) | | (Note 1) | | | | | | | (Note 3) | (Notes 4, 19) |

**PHYSICAL CHARACTERISTICS:**

See figure 1

| | | | |
|---|---|---|---|
| Cooling: | Convection | Mounting position: | Base, horizontal |
| Magnet isolation: | See note 5 | Output coupling: | See figure 1 |
| Mounting support: | See note 6 | Input connections: | See figure 1 |
| | | Weight: | 1 pound approximately (453.59 grams) |

**TEST CONDITIONS:**

| Parameter: | Ef | tk | VSWR |
|---|---|---|---|
| Unit: | V | sec (min) | --- |
| Test condition 1: | 6.3 | 40 | 1.1:1 |
| Test condition 2: | 6.3 | 40 | 1.15:1 |

**GENERAL:**

Qualifications - Required



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| METHOD | REQUIREMENT OR TEST | NOTES | TEST | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MINIMUM | MAXIMUM | |
| | Quality conformance inspection, part 1 | 7 | - | | | | | |
| --- | Transfer loss (1) | 9 | - | F = 120 Hz ±3% | Voltage ratio | 28 | 35 | dB |
| --- | Null reed frequency | 11 | - | | F | 500 | 600 | Hz |
| 1261 | Anode voltage | - | - | Ib = 30 mAdc | Eb | 280 | 330 | Vdc |
| 4250 | Power output (1) | 8,13 | 1 | | Po | 1.0 | --- | W |
| 4223 | Frequency | 14 | 1 | | F | 4,268 | 4,350 | MHz |
| 4250 | Power output (2) | 8,10,13 | 2 | | Po | 1.0 | --- | W |
| --- | FM drive voltage | 15 | 2 | F = 120 Hz ±3% | --- | 0.15 | 0.35 | Vac |
| --- | Distortion | 16 | 2 | | --- | --- | --- | --- |
| 4309 | Dynamic impedance | 19 | - | | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1031 | Low-frequency vibration | - | - | No voltages | --- | --- | --- | --- |
| 4289 | Heater current | - | - | Ef = 6.3 V | If | 0.54 | 0.66 | A |
| 4270 | Heater-cathode leakage | - | - | Ehk = ±45 Vdc | Ihk | --- | 75 | uAdc |
| --- | Reed coils dc resistance | - | - | Stabilized shell, T = 25°C (approx) Coil current = 25 mAdc (max) for test | Drive coil | 0.3 | 0.4 | Ohm |
| | | | | | Pickup coil | 0.55 | 0.7 | Ohm |
| --- | Transfer loss (2) | 9 | - | F = 5,000 Hz ±3% | Voltage ratio | 12 | --- | dB |
| --- | Natural reed frequency | 10 | - | | F | 275 | 350 | Hz |
| --- | Bandwidth linearity | 17 | - | | Deviation | --- | 15 | % |
| 4302 | Frequency change (warmup) | 18 | - | | --- | --- | --- | --- |
| | Room temperature | - | - | After t = 3 minutes | Deviation | --- | ±8 | % |
| | | | | After t = 12 minutes | Deviation | --- | ±8 | % |
| | Cold test (-55°C) | - | - | After t = 10 minutes | Deviation | --- | ±8 | % |
| | | | | After t = 20 minutes | Deviation | --- | ±8 | % |
| | Hot test (+70°C) | - | - | After t = 20 minutes | Deviation | --- | ±4 | % |
| 4310 | Pulling factor | 8,13 | 1 | | ΔF | --- | 4.0 | MHz |
| --- | Reed secondary resonances | 12 | - | | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| --- | Life test | 20 | - | Group C | t | 1,000 | --- | hrs |
| --- | Life-test end points: | | | | | | | |
| 4250 | Power output (1) | 8,13 | 1 | | Po | 0.8 | --- | W |
| --- | Distortion | 16 | - | | Eb | 270 | 330 | Vdc |
| 4309 | Dynamic impedance | 21 | - | | --- | --- | --- | --- |



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NOTES:

1. The maximum voltage applied to the reed drive coil shall not be sufficient to cause distortion of the pickup coil voltage.

2. The maximum value specified herein is for a nonoscillating condition.

3. The temperature shall be measured at the point shown on figure 1.

4. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long" line.

5. In handling and mounting the magnetron, care shall be taken to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within 1 inch of the tube.

6. Nonmagnetic sheet metal clamps shall be used as additional support to that provided by the base pins.

7. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

8. This test shall be performed after at least 20 minutes of tube operation.

9. A rms input of 0.1 Vac shall be applied to the drive coil of the reed modulator at the frequency specified herein. The rms output across the pickup coil terminals shall be within the specified amount with reference to 0.1 Vac. This test shall be made with a high-grade voltmeter.

10. The reed modulator shall be driven over the frequency range specified herein with an input of 0.01 Vac, or less. The frequency shall be adjusted so the rms voltage across the pickup coil terminals is a maximum. The natural reed frequency shall be considered the frequency at which the rms voltage across the pickup coil is a maximum, and shall fall within the frequency limits specified herein.

11. The drive coil of the reed modulator shall be driven over the frequency range specified herein with approximately 0.1 Vac input. The frequency shall be adjusted so the rms voltage across the pickup coil terminals is a minimum. The null frequency shall be considered the frequency at which the rms voltage across the pickup coil is a minimum, and shall fall within the frequency limits specified herein.

12. A constant sinusoidal voltage of 10 millivolts rms shall be applied to the reed drive coil. When the frequency of this voltage is varied from 5 to 10 kHz the voltage across the pickup coil at the secondary resonance shall not increase more than 6 dB above the voltage which would be measured in the absence of the secondary resonance. No secondary resonances shall occur in the range 1 to 5 kHz which is greater than 1 dB above the voltage measured in the absence of the secondary resonance. This test shall be conducted with normal operating voltages applied to the heater and cathode with the tube at normal operating temperature.

13. After the application of Eb, a 2-minute stabilizing period shall be observed. The rate of rise of Eb shall not exceed 50-kilovolts per second.

14. Frequency shall be measured at approximately 30°C, ambient, and at approximately 3 minutes after application of Eb.

15. This voltage shall be obtained from a high-quality audio oscillator. With the reed coils connected in a closed loop circuit as shown on figure 2 and figure 3, or equal, 76 MHz peak-to-peak frequency modulation of the carrier signal shall be obtained with the drive coil voltage adjusted within the limits specified herein. The measurement of rms drive voltage shall be made 6 minutes after the application of Eb, and at least 2 minutes after the application of 76 MHz drive voltage. The closed loop circuit shall be stable. When the audio oscillator voltage is reduced to zero, no voltage indicating closed loop audio frequency oscillation shall be present across the reed modulator drive coil.

16. The tube shall be placed in operation as specified herein for frequency modulation, and the drive voltage shall be increased from 0 volt to 15 percent above that voltage required for 76 MHz peak-to-peak frequency modulation. When viewed on an oscilloscope connected at the magnetron reed drive pins, the waveform of the drive voltage shall not show any visible distortion as compared to the waveform of the audio output voltage of the signal generator. This test shall be made at least 2 minutes after the application of the drive voltage required for 76 MHz of peak-to-peak frequency deviation.



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NOTES: -Continued

17. The peak-to-peak rf bandwidth versus reed drive shall be obtained in the following manner: Plot the 42 and 84 MHz bandwidth versus drive points. A straight line passing through the origin and extending to the 84 MHz point shall be the reference line upon which calculations are based. At the 42 MHz drive level, compute the difference frequency between the actual 42 MHz point and that frequency indicated on the reference line. The calculated difference frequency divided by 42 MHz shall be considered the percentage of nonlinearity, and shall be below the maximum limit specified herein. This test shall be made with the tube stabilized before each measurement.

18. The tube shall be placed in oscillation, as in test condition 2. The peak-to-peak bandwidth at the start of the room temperature test shall be approximately 60 MHz, and the modulator input drive voltage used during the minus 55°C test and the plus 70°C test shall be the same as that used during the room temperature test. In each instance, the bandwidth deviation is with respect to the room temperature test bandwidth at t = 20 minutes. The bandwidth deviation shall stay within the percentage specified herein of the peak-to-peak bandwidth after the times specified herein.

19. The anode current shall be continuously sinusoidally modulated from 5 to 60 milliamperes peak. The tube shall look into an rf system whose VSWR is 2.0 average and which is adjustable in phase. At any load phase, and between the current extremes of 15 and 45 milliamperes, the slope of the voltage current characteristic shall not become less than zero ohms. FM drive voltage shall then be applied at that value previously determined by the test described in note 15. At any load phase there shall be no evidence of moding. The rated heater and anode voltages shall have been applied for at least 4 minutes previous to these tests. The viewing oscilloscope shall be calibrated for approximate sensitivities of 15 milliamperes per inch horizontally and 30 volts per inch vertically when performing this test.

20. Life test shall be conducted with Ef and Eb as specified under conditions for power output (1). A 60 Hz potential of 0.2 Vac shall be applied to the reed drive coil. The pickup coil shall remain open-circuited during this test. Voltages shall be cycled in the following manner:

| Condition | Duration | Ef | Ib | E Reed |
|---|---|---|---|---|
| 1 | 40 sec | 6.3 V | 0 | 0 |
| 2 | 105 min | 6.3 V | 30 mAdc | 0.2 Vac |
| 3 | 15 min | 0 V | 0 | 0 |

Total minimum life time is defined as a total of 1,000 hours of condition 2.

21. The anode current shall be continuously modulated from 5 to 60 milliamperes peak. The tube shall look into a mismatched load whose VSWR is 1.50 minimum and which is adjustable in phase. At any load phase and between the current extremes of 15 and 45 milliamperes, there shall be no break in the voltage-current characteristic.

22. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 19

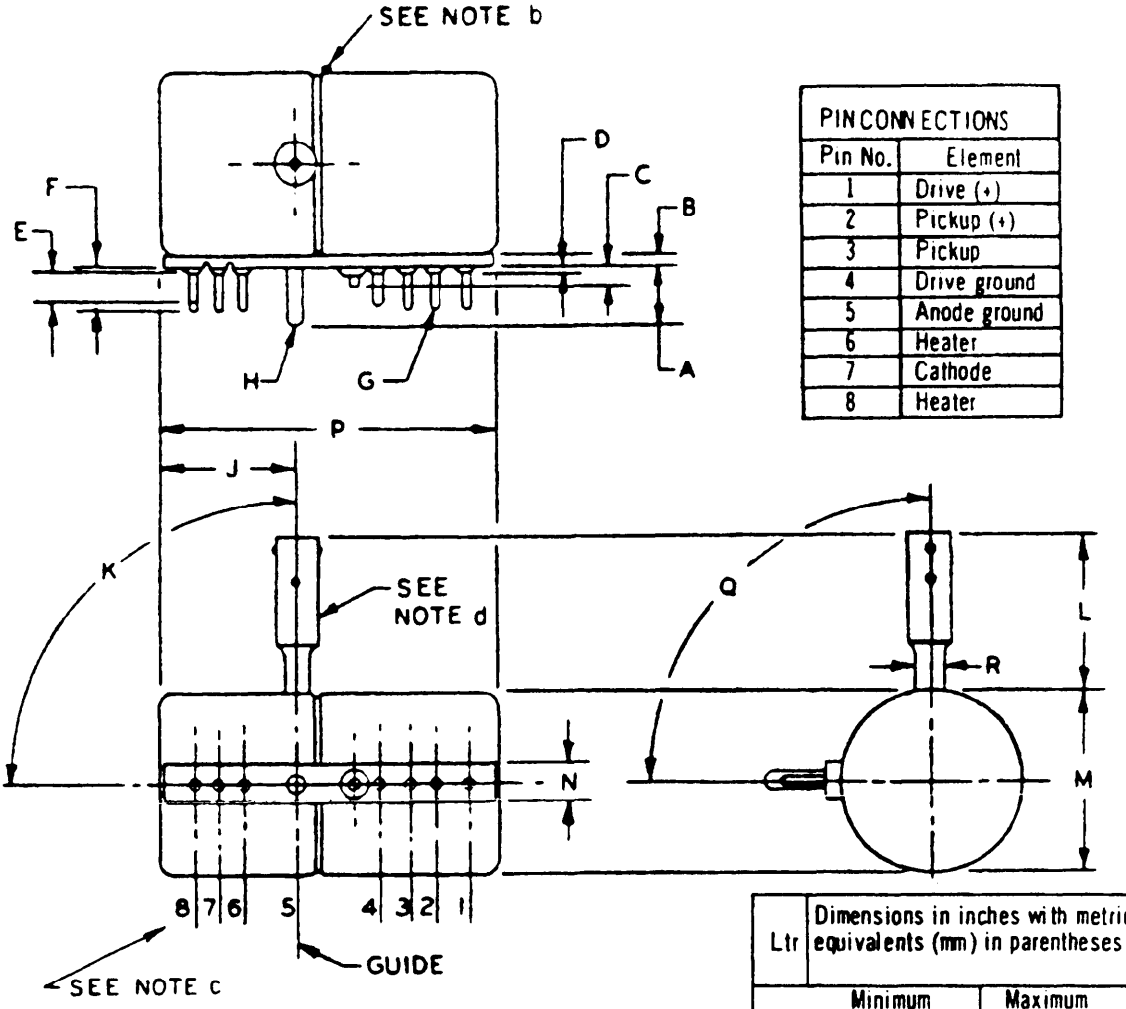
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2893)



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| PIN CONNECTIONS | |
|---|---|
| Pin No. | Element |
| 1 | Drive (+) |
| 2 | Pickup (+) |
| 3 | Pickup |
| 4 | Drive ground |
| 5 | Anode ground |
| 6 | Heater |
| 7 | Cathode |
| 8 | Heater |

NOTES:

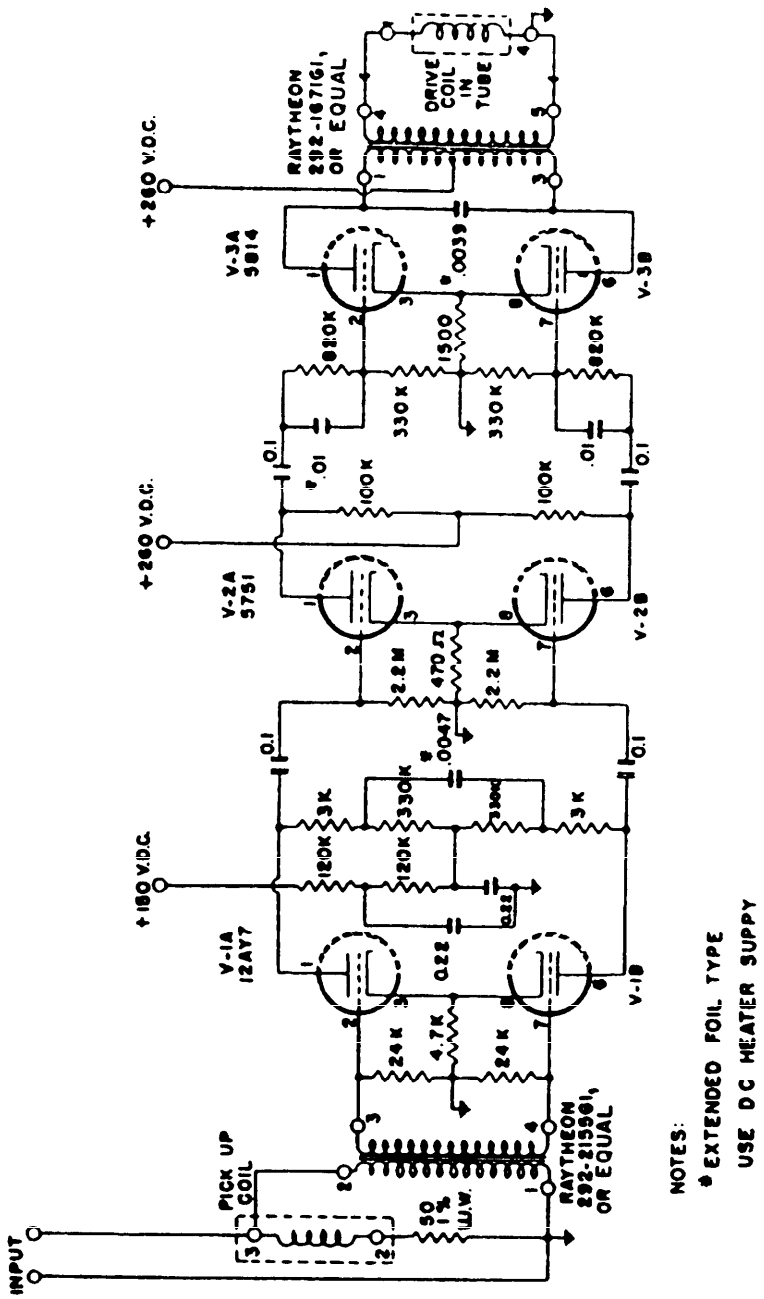
a. Tube base must be capable of being checked in accordance with the following procedure: The base pin gage (see figure 4) shall be attached to such an accessory weight as will give a total weight of 2 pounds. The entire length of pin shall pass into the gage and on withdrawal shall become disengaged without lifting the total weight of assembled gage and accessory weight.
b. Anode temperature to be measured at this point.
c. The cathode shall be connected to pin No. 7 through a 1,000-ohm resistor enclosed within the tube shell.
d. Couple with BNC connector M39012/17-0001 or equal.

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 1 | | |
| K | 85° | 95° |
| M | | 1.656 (42.06) |
| Q | 85° | 95° |
| Quality conformance inspection, part 2 | | |
| H | .120 (3.05) | .130 (3.30) |
| L | 1.344 (34.14) | 1.460 (37.08) |
| P | | 3.250 (82.55) |
| Quality conformance inspection, part 3 | | |
| A | | .400 (10.16) |
| C | | .125 (3.17) |
| D | | .050 (1.27) |
| E | .190 (4.83) | |
| F | | .300 (7.62) |
| G | .061 (1.550) | .066 (1.676) |
| J | 1.234 (31.34) | 1.358 (34.49) |
| R | | .275 (6.98) |
| | Nominal dimensions | |
| B | .125 (3.18) | |
| N | .312 (7.92) | |

FIGURE 1. Outline drawing of electron tube type 6177A.



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NOTES:
* EXTENDED FOIL TYPE
USE DC HEATER SUPPY

FIGURE 2. Reed modulator circuit.



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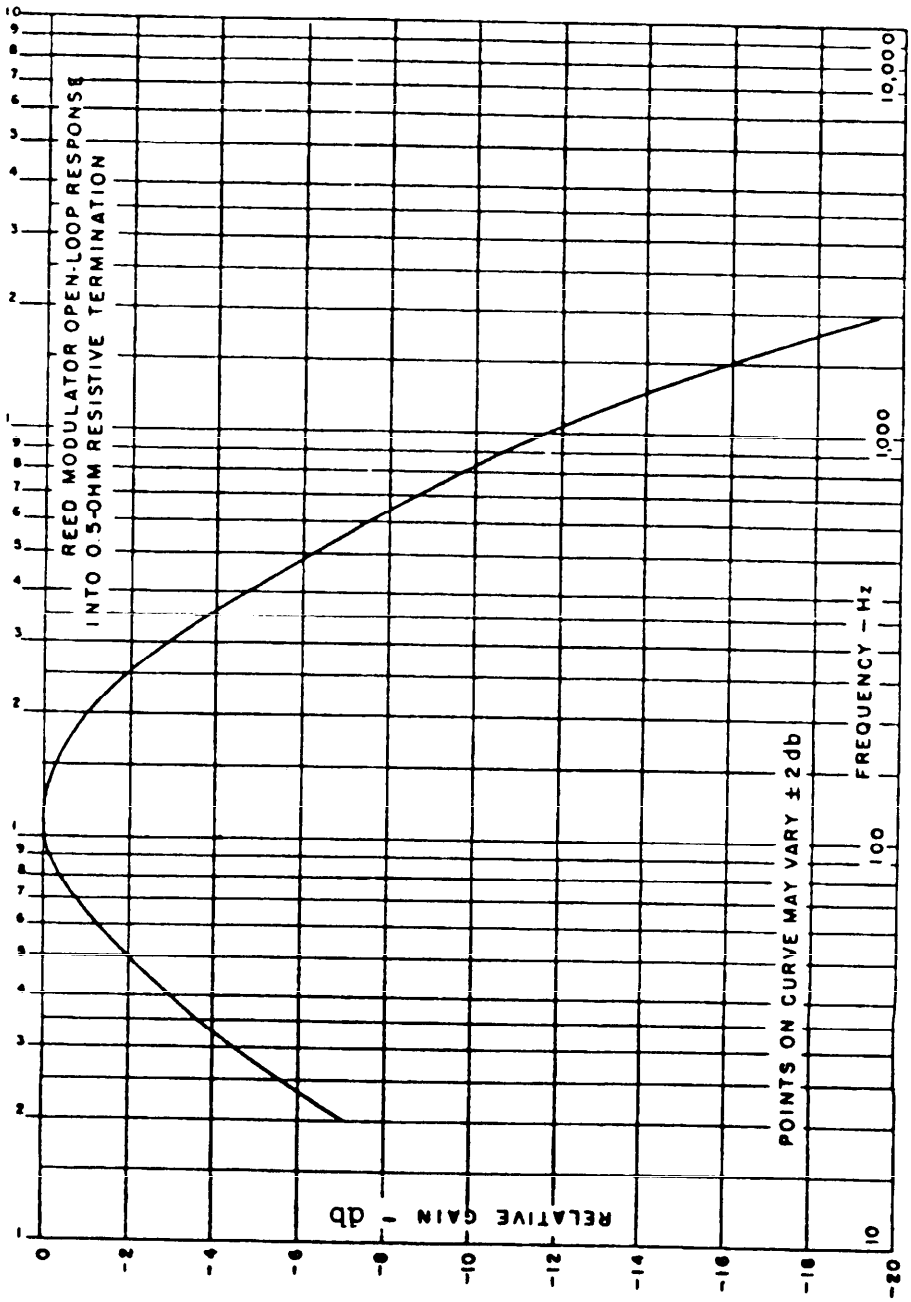
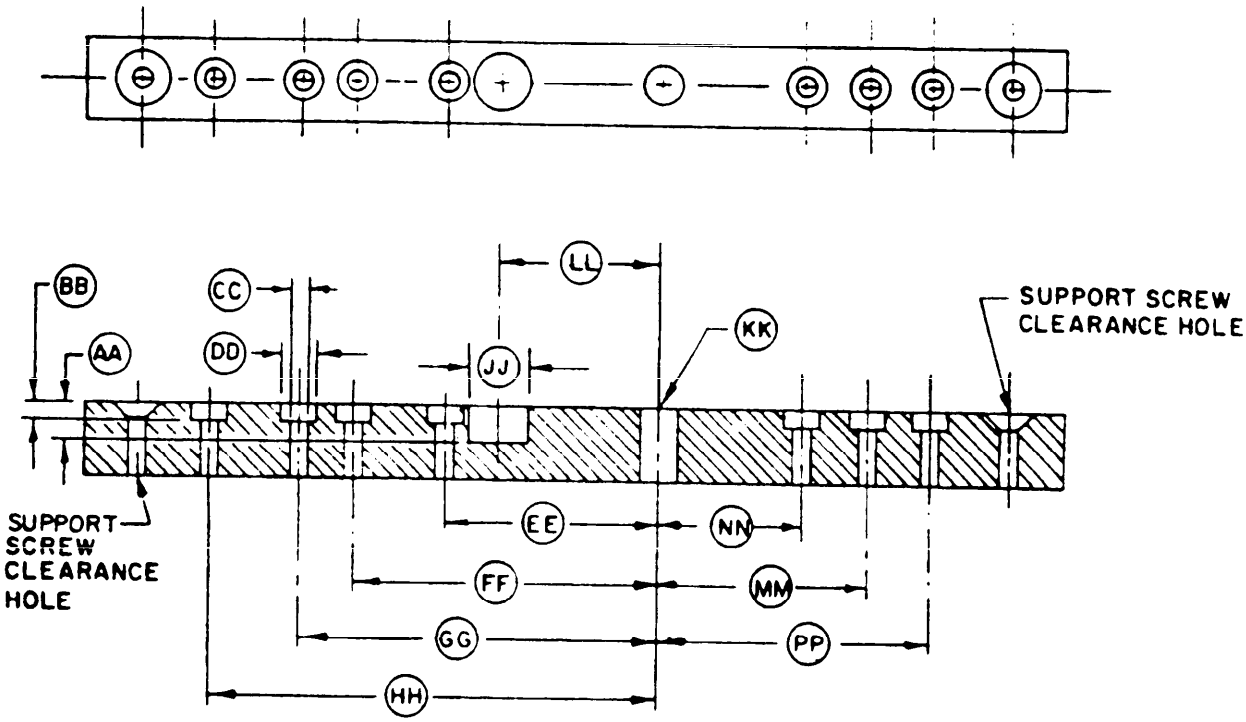



FIGURE 3. Response curve.



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| | Limits | | |
|---|---|---|---|
| | Inches | | |
| | Min | Max | Nom |
| AA | .124 | .126 | .125 |
| BB | .061 | .063 | .062 |
| CC | .076 | .077 | |
| DD | .119 | .121 | .120 |
| EE | .755 | .757 | .756 |
| FF | 1.037 | 1.039 | 1.038 |
| GG | 1.263 | 1.265 | 1.264 |
| HH | 1.551 | 1.553 | 1.552 |
| JJ | .234 | .236 | .235 |
| KK | .128 | .130 | |
| LL | .546 | .548 | .547 |
| MM | .715 | .717 | .716 |
| NN | .490 | .492 | .491 |
| PP | .940 | .942 | .941 |

FIGURE 4. Base pin gage. (SEE NOTE o)



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